

DATA PREPROCESSING OF DIRTY CAFÉ SALES DATA

1. Introduction

In today's food and beverage industry, businesses increasingly rely on data to guide their decisions. Using data effectively helps companies improve their daily operations, understand what customers want, and increase profits. Every time a sale is made, valuable data is created. When this data is analysed properly, it can reveal useful patterns, such as which products are most popular, when sales peak, and which payment methods customers prefer.

This report examines a dataset containing cafe sales records. The dataset is intentionally dirty, meaning it includes missing values, inconsistencies, and errors. These issues reflect the types of challenges often encountered with real-world data, where information may be recorded incorrectly due to human error, system failures, or inconsistent data entry methods. Dirty data is common in practical settings, and handling it properly is essential before building reliable machine learning models. If these data quality problems are ignored, they can lead to inaccurate results, poor model performance, and misleading conclusions. The main goal of this analysis is to clean and prepare the raw data so that it can be used for future predictive modelling tasks.

2. Materials and Methods

2.1 Data Source

The dataset used for this analysis is the Dirty Cafe Sales Dataset, obtained from the Kaggle data science platform.

Source: Kaggle

Link to the Dataset: <https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>

This dataset is appropriate for this study for multiple reasons. It contains about 10,000 records and 8 features, which offer enough complexity for meaningful analysis. It also includes a mix of data types, numerical values such as whole numbers (Quantity) and decimals (Price Per Unit, Total Spent), as well as categorical or text-based data (Item, Payment Method, Location, Transaction Date). Most importantly, the dataset is intentionally dirty, meaning it includes realistic data quality problems like missing entries, placeholder values such as "UNKNOWN," and clear error indicators like "ERROR." These characteristics are common in real-world transactional data, making this dataset ideal for practicing the core concepts of identifying missing data, understanding the reasons behind it, and applying imputation methods. The variety of data types also makes it possible to apply encoding techniques that are widely discussed in modern data science literature.

2.2 Data description and exploratory data analysis

Observed Patterns and Trends

Quantity Purchased:

The most common quantities are 2 and 3 items per transaction, each occurring over 1,750 times. The distribution is spread evenly across values 1 to 5, with no single quantity dominating.

Total Spent:

The distribution of Total Spent is right-skewed, meaning most transactions are small between 4\$ - 8 \$, with fewer large transactions reaching up to 25 \$. This is typical in retail settings.

Price Per Unit:

Prices are clustered at 2, 3, 4, and 5 \$, which correspond to different product types for example., coffee at 2 \$, sandwiches at 4 \$.

Table 1: Correlations Between Variables:

	QUANTITY	PRICE PER UNIT	TOTAL SPENT
QUANTITY	1	0.00	0.68
PRICE PER UNIT	0.00	1	0.70
TOTAL SPENT	0.68	0.70	1

Total Spent is strongly linked to both Quantity ($r = 0.68$) and Price Per Unit ($r = 0.70$). This makes sense, as total spend is calculated from these two values.

There is no relationship between Quantity and Price Per Unit ($r = 0.00$). Customers are just as likely to buy multiple low-cost items as multiple high-cost items.

Transaction Dates:

Transactions span the full year of 2023, with a fairly even spread across months.

Outliers and Anomalies

Total Spent has a maximum of 25 \$, which is mathematically correct for example, 5 salads at 5\$ each. These high values are legitimate but appear as outliers because they are less common.

Categorical columns contained "ERROR" and "UNKNOWN" values, which were treated as missing data.

Numerical values are all within expected ranges, indicating that existing data is reliable.

Table 2: Data Quality Challenges

CHALLENGE	DESCRIPTION	IMPACT ON MODELLING
MISSING VALUES	Several columns have missing data	Must be handled to avoid biased results

PLACEHOLDER VALUES	"ERROR" and "UNKNOWN" in categorical columns	Need to be treated as missing
SKEWED TARGET	Total Spent has a right-skewed distribution	May affect model performance; robust methods needed
STRONG CORRELATION	Total Spent is highly correlated with predictors	Expected, but may cause issues in some linear models

2.3 Identifying the Missing Data Mechanism

Based on the missing data analysis, the mechanism is classified as Missing Completely At Random (MCAR) .

Evidence Supporting MCAR Classification:

Correlation of Missingness: The correlation matrix of missing indicators shows near-zero correlations across all columns:

	PAYMENT METHOD	LOCATION	TRANSACTION DATE
PAYMENT METHOD	1.000000	0.014792	0.016339
LOCATION	0.014792	1.000000	-0.001088
TRANSACTION DATE	0.016339	-0.001088	1.000000

These extremely low correlations (all < 0.02) indicate that missingness in one column is not systematically related to missingness in another column. This pattern is characteristic of MCAR, where missing values occur independently across variables (Sadegh et al., 2025).

The data satisfies the definition of MCAR because the probability of missingness is independent of both observed and unobserved data. The statistical evidence from correlation analysis and the random visual pattern all converge to support this classification. This has important implications for handling strategies: MCAR allows for the widest range of imputation techniques without introducing systematic bias, though listwise deletion would also produce unbiased estimates with reduced sample size.

2.4 Data Encoding

Two encoding techniques were applied to transform categorical variables into numerical formats suitable for machine learning algorithms.

One-Hot Encoding for Payment Method

One-hot encoding was applied to the Payment Method variable, creating four binary columns: payment_method_cash, payment_method_credit, payment_method_digital_wallet, and payment_method_missing. One-hot encoding is the appropriate choice for Payment Method

because it is nominal categorical data with no inherent order or ranking among categories . The categories: Cash, Credit Card, and Digital Wallet, represent distinct payment types, but none is greater than or superior to another. One-hot encoding keeps this nominal nature by creating independent binary indicators for each category, preventing any machine learning algorithm from assuming false ordinal relationships.

Frequency Encoding for Item

Due to the high cardinality of the Item variable (8 unique categories), frequency encoding was applied instead of one-hot encoding.

Frequency encoding was selected for the Item variable because one-hot encoding with 8+ categories would create an excessive number of new columns, increasing dimensionality and potentially leading to the curse of dimensionality in downstream modelling. Frequency encoding replaces each item with its relative frequency in the dataset, capturing the popularity of each item while maintaining a single numerical feature.

2.5 Imputation Techniques

Method 1: Median Imputation for Numerical Variables

Median imputation was used for numerical variables (Quantity, Price Per Unit, Total Spent) because it is robust to outliers. As observed in the EDA section, Total Spent contains outliers on the higher end of its distribution. Using mean imputation would make more sense to these values, potentially pulling imputed values away from the centre of the typical transaction distribution. The median preserves the central tendency without being influenced by outliers.

Method 2: Mode Imputation for Categorical Variables

Mode imputation was used for the categorical variables after they were encoded. For the one-hot encoded payment method columns, missing values were filled with 0, which is appropriate since these are binary indicators where missing implies the category is not present. For categorical variables in general, mode imputation (replacing missing values with the most frequent category) is the standard approach because it preserves the most common class without introducing arbitrary values.

Method 3: Regression Imputation

Regression imputation was used as another approach. This method uses the relationship between variables to predict missing values. Given the strong correlation between Price Per Unit and Total Spent ($r = 0.70$) and between Quantity and Total Spent ($r = 0.68$), regression imputation is appropriate for this dataset. Missing Total Spent values can be predicted from known Quantity and Price Per Unit values using a linear regression model. This approach preserves the mathematical relationship between variables and generally produces more accurate imputations than simple univariate methods.

Implications of Imputation on Data Integrity

The way we handle missing data can affect the quality of the dataset, introduce bias, and influence the results of any models built later. Each imputation method has its own strengths and weaknesses.

Median Imputation:

Using the median to fill missing values keeps the data centred around a typical value and is not affected by extreme outliers. This is useful because Total Spent has some very high values. However, a downside is that repeating the same value for many missing entries reduces the variance of the data. This can make relationships between variables appear weaker than they really are and may lead to models that are overconfident in their predictions.

Mode Imputation:

For categorical data, filling missing values with the most common category keeps the original distribution intact. But when many values are missing like with Payment Method (31.76%), this approach can over-represent the most frequent category and introduce bias. To reduce this problem, a 'Missing' category was created before encoding, and missing values in the binary columns were set to 0. This way, the fact that a value was missing is preserved, and the model can still learn if missingness itself follows any patterns.

Regression Imputation:

Regression imputation uses the relationships between variables to make smarter guesses about missing values. For example, if a transaction has Quantity = 4 and Price Per Unit = 5.0, the missing Total Spent can be estimated as around 20\$, which is mathematically correct and matches similar transactions. This method usually produces more realistic imputations because it considers how variables are connected. However, it assumes that the relationships in the data are linear, which may not always be true.